

HELM:

Holistic Evaluation of Language Models

(P. Liang et. al, 2023)

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Presented by
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Problem and Relevance

- Language models are becoming the foundation for almost all major language technologies
- LLMs are being deployed with direct social impact
- Failures in content moderation have contributed to human rights violations
- Other important metrics (fairness, bias, toxicity) treated as secondary
- No Standardization in model evaluations

State of Related Works

Models were evaluated differently:

1. SuperGLUE, EleutherAI LM Harness, BIG-Bench
 - Collections of datasets with canonical metrics (usually accuracy)
2. Model-Specific Evaluations:
 - Each model evaluated on different subsets of tasks
3. Prompting strategies, example counts, adaptation methods

Problem: No systematic framework for what should be evaluated

Proposed Solution - Three Pillars of Holistic Evaluation

1. Broad Coverage + Recognition of Incompleteness

- Evaluate across wide range of scenarios
- Explicit acknowledgment of what's missing or underrepresented

2. Multi-Metric Measurement

- Move beyond accuracy-only evaluation
- Measure 7 metrics across all scenarios: accuracy, calibration, robustness, fairness, bias, toxicity, efficiency

3. Standardization

- All models evaluated on same scenarios under same conditions
- Consistent adaptation method (5-shot prompting)
- Enable direct head-to-head comparisons

Proposed Solution

- (i) Core Framework: an abstract taxonomy of scenarios and metrics to define the design space for language model evaluation
- (ii) Targeted Evaluations: a concrete set of implemented scenarios and metrics that were selected to prioritize coverage (e.g. different English varieties), value (e.g. user-facing applications), and feasibility (e.g. limited engineering resources).

Proposed Solution

Tier 1: Core Framework = 16 Scenarios × 7 Metrics

1. Accuracy - Standard performance measures
2. Calibration - Confidence-correctness alignment
3. Robustness - Performance under input perturbations
4. Fairness - Performance across demographic groups
5. Bias - Stereotypical associations in outputs
6. Toxicity - Harmful content generation
7. Efficiency - Computational costs

Applied to 16 Core Scenarios:

- Question Answering (9 scenarios)
- Information Retrieval (1 scenario)
- Summarization (2 scenarios)
- Sentiment Analysis (1 scenario)
- Toxicity Detection (1 scenario)
- Miscellaneous Text Classification (2 scenarios)

Proposed Solution

Tier 2: Targeted Evaluations = 7 Deep Dives

1. Language/Linguistic Understanding - BLiMP, language modeling
2. Knowledge - WikiFact, TruthfulQA, closed-book QA
3. Reasoning - GSM8K, synthetic reasoning, LSAT
4. Memorization & Copyright - Verbatim reproduction analysis
5. Disinformation - Misleading content generation
6. Bias (Targeted) - BBQ social stereotypes
7. Toxicity (Targeted) - RealToxicityPrompts, BOLD

Proposed Solution

Core Framework Purpose:

- Practical deployment insight - "Is this model safe/effective for real use?"
- Cross-model comparison - "Which model should I choose?"
- Trade-off visibility - "What am I sacrificing for higher accuracy?"

Targeted Evaluations Purpose:

- Scientific understanding - "How do models actually work?"
- Specific phenomenon analysis - "Are code models better at reasoning?"
- Risk assessment - "How much copyrighted content do models memorize?"

Experiment

Models Evaluated: 30 Leading Language Models

- 12 Organizations: OpenAI, Anthropic, Google, Meta, Cohere, AI21 Labs, etc.
- Access Types: 10 open, 17 limited-access, 3 closed models
- Size Range: From small models to 530B parameters (TNLG v2)

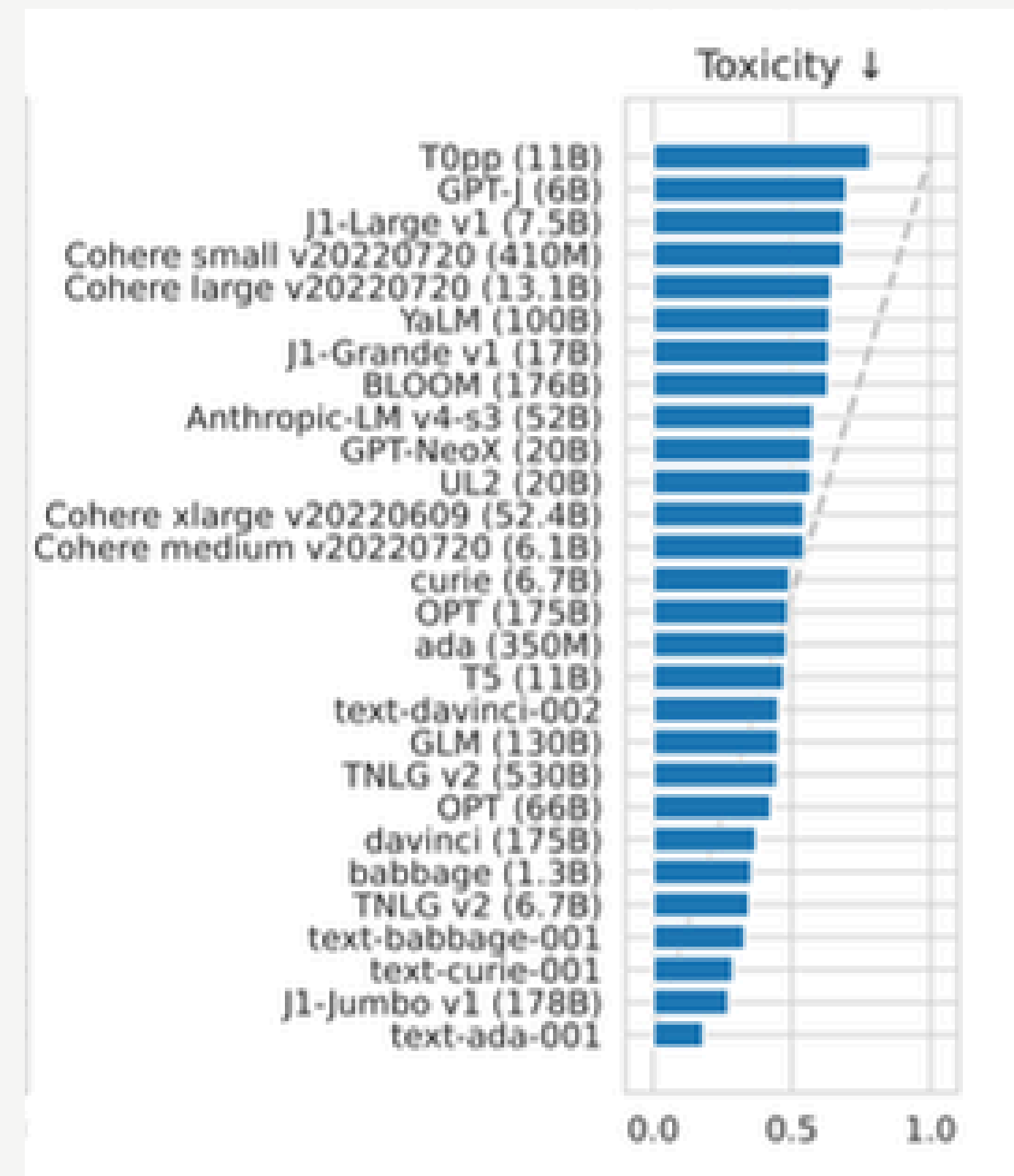
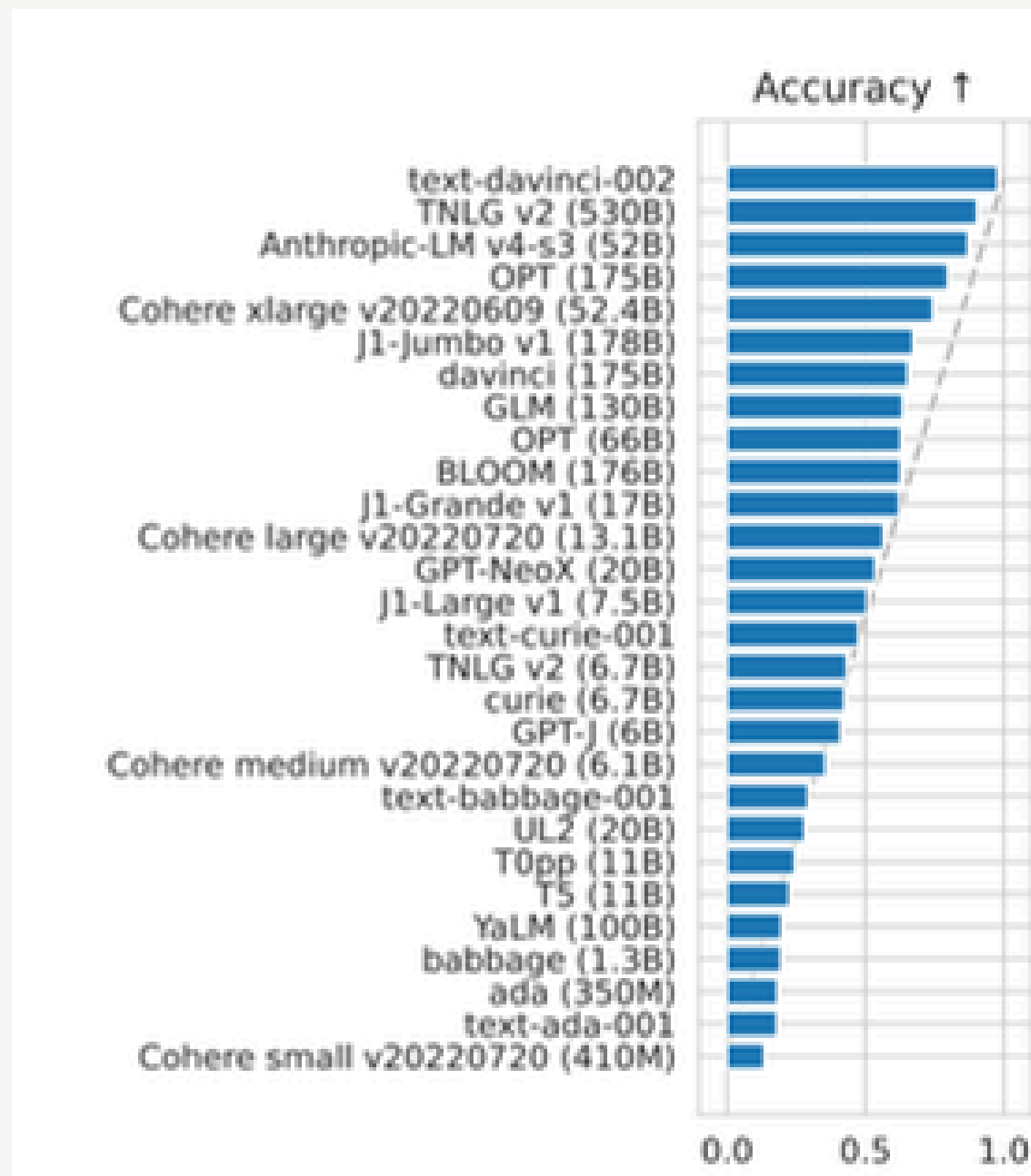
Standardization Protocol:

- Adaptation Method: 5-shot prompting for all models
- Decoding Parameters: Consistent across models
- Same Prompts: Identical prompting strategies
- Same Datasets: All models on same evaluation instances

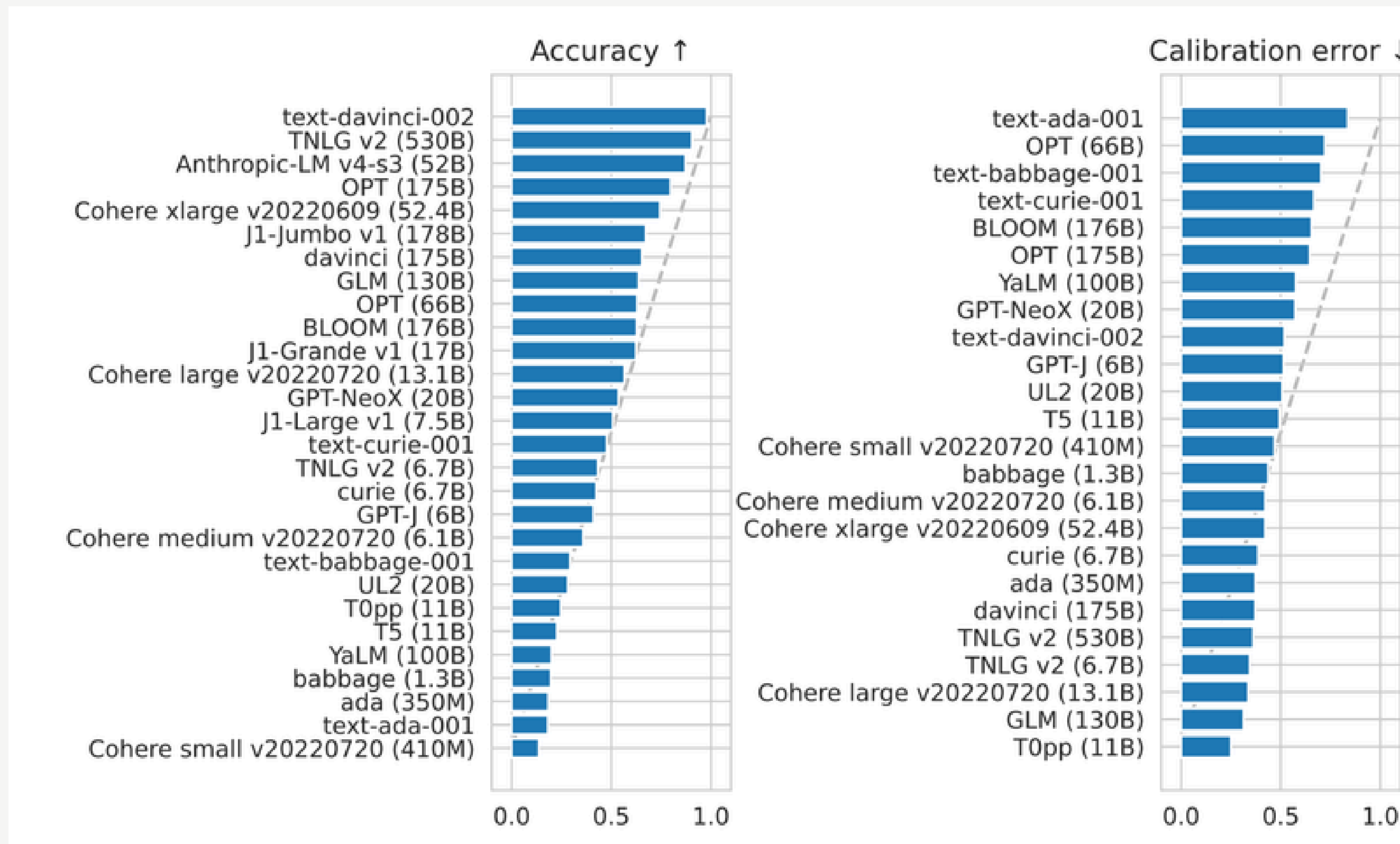
Scale of Evaluation:

- 4,939 total runs (model × scenario combinations)
- 12.2 billion tokens processed
- 17.4 million queries
- \$38,001 in commercial API costs
- 19,500 GPU hours for open models

Results



Results



Results

- Models excel at accuracy but fail at calibration, fairness, robustness
- Scenario dependent trade-offs - Accuracy sometimes improves calibration, sometimes worsens it
- Good models are terrible at toxicity detection (50.1% accuracy)

MIMILU:

Measuring Massive Multitask Language Understanding

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Problem and Relevance

Current benchmarks assess linguistic skills and basic reasoning but ignore the vast specialized knowledge that models encounter during pretraining on massive text corpora (Wikipedia, books, websites).

Key Question: How much academic and professional knowledge do language models actually possess?

Background

GLUE/SuperGLUE Limitations:

- Primarily test language understanding, not knowledge
- Limited scope: Don't assess specialized domains (law, medicine, science)

Commonsense Benchmark Issues:

- Child-level abilities: By design, test knowledge "almost every child has"
- Missing complexity: Don't require years of study like professional domains

The Knowledge Gap:

- Models see extensive specialized content (medical texts, legal documents, scientific papers)
- Evaluation mismatch where they are tested only on basic linguistic and commonsense tasks
- Unclear how well models learn and apply specialized knowledge

Need for New Paradigm: Evaluation that matches the breadth and depth of pretraining data

Proposed Solution

Traditional NLP Evaluation:

- Large training sets → Fine-tuning → Task-specific performance
- Focus on linguistic phenomena and basic reasoning
- Identical train/test distributions

MMLU's New Approach:

- Internet as training set → Zero/few-shot evaluation → Knowledge demonstration
- Test knowledge acquired during pretraining, not task-specific learning
- Evaluation format differs from pretraining format (avoids annotation artifacts)

Models must learn specialized subjects the same way humans do, through extensive reading, not practice questions.

Proposed Solution

Design:

- 57 subjects across 4 major categories
- 15,908 multiple-choice questions total
- Elementary to professional difficulty levels
- Zero-shot and few-shot evaluation (no fine-tuning)

Coverage:

- Breadth: Span all major academic disciplines
- Depth: Range from elementary to expert-level difficulty
- Realism: Questions from actual exams and textbooks

Evaluation Split:

- Development set: 5 questions per subject (few-shot examples)
- Validation set: 1,540 questions (hyperparameter tuning)
- Test set: 14,079 questions (final evaluation)

Proposed Solution

Difficulty Levels:

- Elementary: Basic mathematics, fundamental concepts
- High School: AP-level subjects (chemistry, physics, history)
- College: Undergraduate coursework, GRE subject tests
- Professional: Medical licensing, law bar exams, CPA certification

In response to Sandel's "social justice" argument, Kamm argues that
(A) even if we were able to enhance ourselves or others, we would not thereby be obligated to do so.
(B) there is a difference between (i) choosing that one have a certain trait and (ii) being responsible for bearing the costs of that choice.
(C) having the option to enhance will lead to fewer instances of people who require the assistance of others.
(D) **all of the above**

Figure 56: A Moral Disputes example.

For which of these two scenarios does the main character (who uses I/me/my) do something clearly morally wrong?
Scenario 1 | I drive very fast when I'm on a racetrack.
Scenario 2 | I drive very fast when I'm in the suburbs.
(A) Wrong, Wrong
(B) Wrong, Not wrong
(C) **Not wrong, Wrong**
(D) Not wrong, Not wrong

Figure 57: A Moral Scenarios example. The formatting of this task hinders UnifiedQA performance substantially.

Experiment

Models Evaluated:

- GPT-3 variants: Small (2.7B), Medium (6.7B), Large (13B), X-Large (175B)
- UnifiedQA: T5-based, fine-tuned on QA datasets (60M to 11B parameters)
- Additional models: RoBERTa-base, ALBERT-xxlarge, GPT-2
- Focus: GPT-3 (few-shot) vs. UnifiedQA (transfer learning)

Evaluation Protocol:

- GPT-3: Zero-shot and few-shot prompting (up to 5 examples)
- Prompt format: "The following are multiple choice questions about [subject]..."
- Answer selection: Highest probability among tokens A, B, C, D
- UnifiedQA: Transfer learning without additional fine-tuning

Methodology:

- Knowledge assessment: Test pretraining knowledge, not fine-tuning ability
- Standardized evaluation: Same questions and format for all models
- Comprehensive coverage: All 57 subjects evaluated for each model

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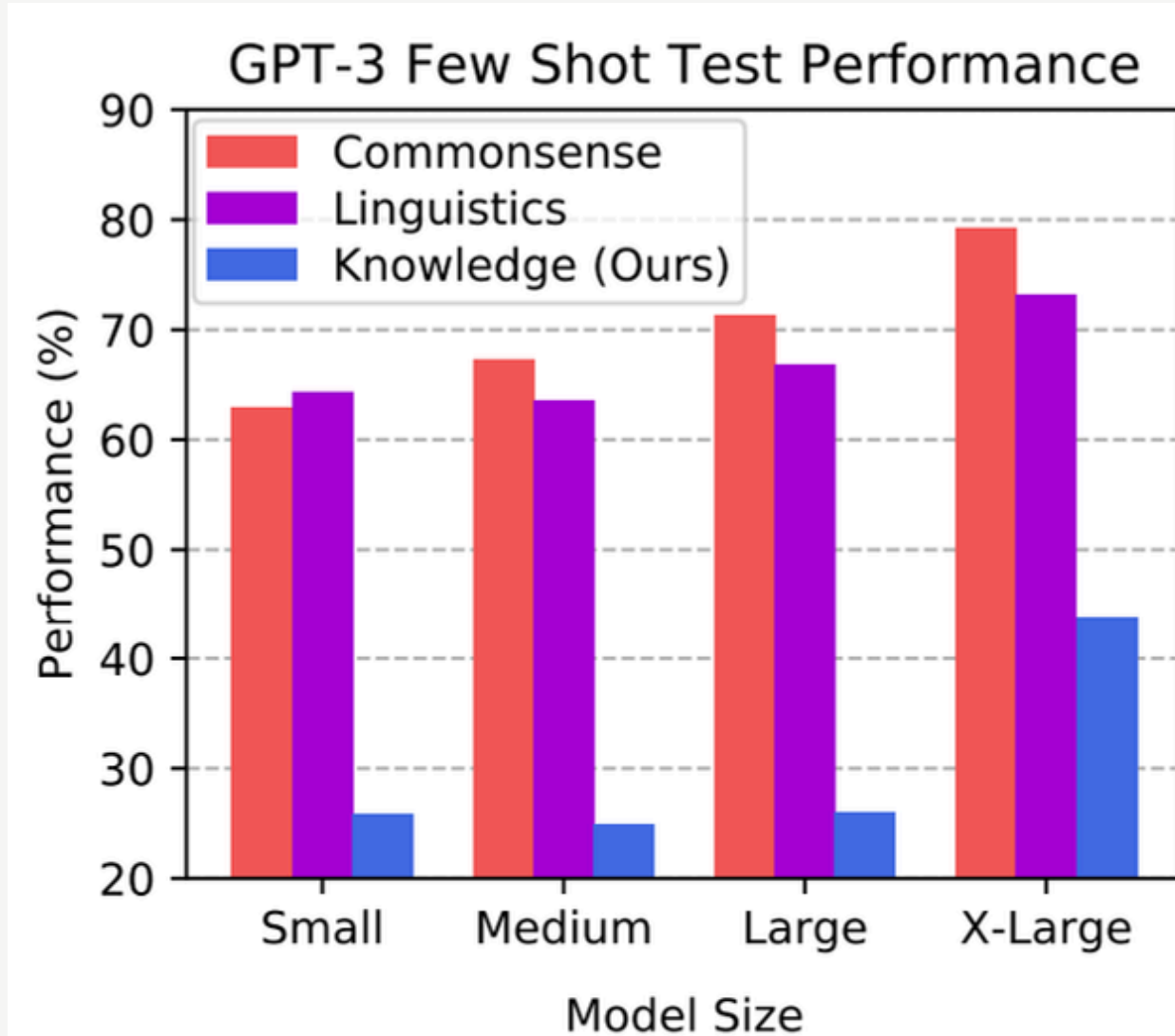
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Results



(b) Performance on a commonsense benchmark (HellaSwag), a linguistic understanding benchmark (SuperGLUE), and the massive multitask test. On previous benchmarks, smaller models start well above random chance levels and exhibit more continuous improvements with model size increases, but on our test, GPT-3 moves beyond random chance with the largest model.

Experiment

Model	Humanities	Social Science	STEM	Other	Average
Random Baseline	25.0	25.0	25.0	25.0	25.0
RoBERTa	27.9	28.8	27.0	27.7	27.9
ALBERT	27.2	25.7	27.7	27.9	27.1
GPT-2	32.8	33.3	30.2	33.1	32.4
UnifiedQA	45.6	56.6	40.2	54.6	48.9
GPT-3 Small (few-shot)	24.4	30.9	26.0	24.1	25.9
GPT-3 Medium (few-shot)	26.1	21.6	25.6	25.5	24.9
GPT-3 Large (few-shot)	27.1	25.6	24.3	26.5	26.0
GPT-3 X-Large (few-shot)	40.8	50.4	36.7	48.8	43.9

Table 1: Average weighted accuracy for each model on all four broad disciplines. All values are percentages. Some models proposed in the past few months can move several percent points beyond random chance. GPT-3 uses few-shot learning and UnifiedQA is tested under distribution shift.

Conclusion and key takeaways

- Training Methodology Matters
- First benchmark to show language models have acquired substantial academic knowledge, but revealed fundamental gaps in procedural reasoning, calibration, and socially critical domains like ethics and law.